AI-BASED EARLY CHILDHOOD EMOTIONAL INTELLIGENCE MONITORING PLATFORM IN URBAN AREAS

Project Id: 25-26J-088

Project Proposal Report

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Sri Lanka Institute of Information Technology
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DECLARATION

We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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The above candidate is carrying out research for the undergraduate Dissertation under my supervision.

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27/08/2025

Date

ACKNOWLEDGEMENT

I would like to extend my sincere gratitude to my supervisor, Dr. Lakmini Abeywardhana, and my co-supervisor, Mrs. Lokesha Weerasinghe, from the Faculty of Computing, for their valuable advice and assistance at the beginning of my research project. I also wish to express my appreciation to our external supervisor, Dr. Naren Deepan, for his mentorship and insightful guidance on the psychological aspects of this study.

Their constructive feedback and critical suggestions have greatly enhanced my focus and contributed to improving the overall quality of my research. I would also like to acknowledge the contributions of my fellow team members for their continuous support and collaboration throughout this journey.

Finally, I want to express my gratitude to everyone who volunteered to assist in this research. Sharing their experiences and viewpoints has been essential in helping me develop a deeper understanding of the topic.

ABSTRACT

This research proposes the design of an Artificial Intelligence (AI) driven memory recall assessment module to support the monitoring of Emotional Intelligence (EI) in preschool children aged 4-6 years in urban, Sri Lanka. Early childhood is a critical stage for cognitive and socio-emotional development, yet existing recall assessment tools face multiple limitations: they are overly reliant on screen-based platforms, emphasize motor performance rather than actual memory retrieval, lack multimodal integration, and are designed almost exclusively for alphabetic languages such as English. These gaps are particularly evident in bilingual environments like Sri Lanka, where children must recall and reproduce both the visually complex Sinhala alphasyllabary and the simpler English alphabetic script. To address this, the proposed study develops a multimodal framework that combines handwriting recall tasks in Sinhala and English with categorization-based tasks involving familiar objects such as fruits, vegetables, and animals. Teachers will administer these tasks in classrooms, and completed responses will be uploaded for AI-based analysis using computer vision and deep learning models. A weighted scoring approach, informed by background survey data, will ensure fairness across script complexities and socio-cultural factors. The outputs will be presented through a teacher-facing dashboard with diagnostic insights and parent-facing progress reports, enabling both early intervention and continuous monitoring of self-regulation, a key EI construct. The anticipated outcome is a culturally responsive, non-screen dependent, and scalable assessment framework that directly measures recall rather than proxies, integrates multimodal evidence of memory performance, and strengthens collaboration between teachers and families. By bridging critical methodological and contextual gaps, this research aims to contribute a novel tool for early childhood assessment that balances technological innovation with developmental appropriateness.

Keywords: Memory recall, handwriting, categorization, emotional intelligence, self-regulation

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LIST OF ABBREVIATIONS

AI - Artificial Intelligence

CLIP - Contrastive Language-Image Pre-training

CNN - Convolutional Neural Network

CTC - Connectionist Temporal Classification

EI - Emotional Intelligence

GPU - Graphics Processing Unit

LLM - Large Language Model

MDRNN - Multidimensional Recurrent Neural Network

OCR - Optical Character Recognition

SSD - Single Shot Multibox Detector

ViT - Vision Transformers

YOLO - You Only Look Once

1. INTRODUCTION

Emotional Intelligence, usually described as the ability to recognize, understand, manage, and use emotions effectively in oneself and others [1]. By developing emotional intelligence, people learn more about themselves and learn to cooperate with those around them [2]. For early childhood, emotional intelligence is crucial because lower emotional intelligence impacts clear thinking process and the education of preschoolers [3].

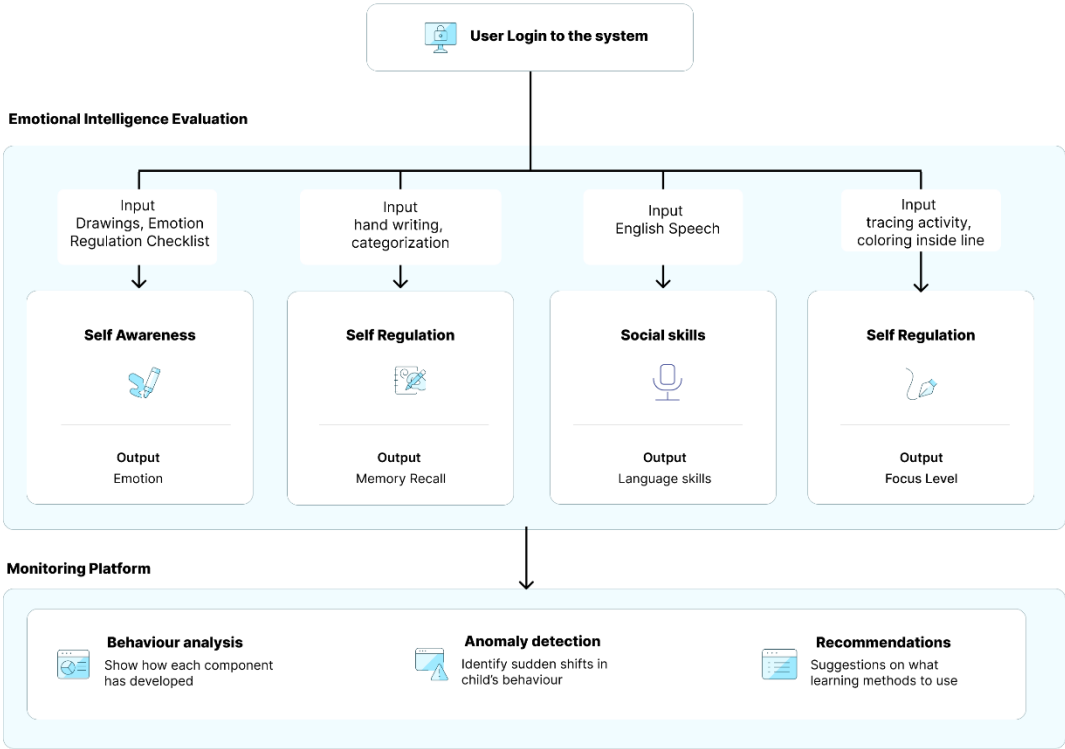

Figure 1.1: System Overview Diagram

EI has been speculated to be a leading factor as to why children are well adapted in schools, their social adaptation and lifelong wellbeing. Described by Goleman as having the ability to perceive, understand and manage emotions, EI has five pillars including self-awareness, self-regulation, motivation, empathy, and social skills [4]. Within them, self-regulation plays a leading role in early childhood since it gives children the ability to adjust, time-delay and calmly react to adverse situation [5].

Memory recall is a mental resource in self-regulation as it provides children with an opportunity to access knowledge in a set of problem-solving and/or emotional situations. Cued recall is more valid during estimation of long-term memory than the free recall, particularly in 4-6-year-old children [6]. This is in line with the levels of processing framework by Craik and Lockhart in that greater semantic processing of information leads to superior retention and recall [7]. To preschoolers, this implies that memory recall tasks within the context of meaningful activities like handwriting or categorization is a quite effective method of measuring the performance of memory.

Handwriting provides a natural and strong context upon which one retrieves memory. Neuroscience experiments have found that writing and handwriting in particular stimulates multiple brain networks involved in vision, motor-planning and memory, enhancing retention beyond forms of input that do not engage the hands (screen-based activities and typing in particular) [8], [9]. Bilingual contexts, like Sinhala and English learning in urban preschools in Sri Lanka, also incur extra difficulties and potentialities: recall accuracy in the two writing systems gives special insights into memory development [10]. To supplement handwriting, other tasks may involve categorization (e.g., classifying animals or objects) which measures one of the most important components of language and cognitive development semantic memory organization [11],[12].

Current child assessment products are also screen-intensive and can lead to issues of digital over-exposure, as well as a form of reduced authenticity in interaction [13]. This proposal will fill that gap by designing a multimodal, bilingual memory recall module that integrates into an AI-based EI monitoring system that is screen-free. The given module enables assessing long-term memory in children through mixing handwriting skills and the task of categorizing the studied material in a way that is most directly applicable to tracking EI, specifically, the pillar of self-regulation.

1.1. Background and Literature Survey

1.1.1. Background

We carried out an online survey to find out how important it is to keep track of emotional intelligence in early childhood. This survey was conducted by sending the

online form to parents and the teachers in urban areas in Western Province. The survey results will help in building the application in a more successful manner. The form was filled by 64 individuals. Below are the results of the form.

Figure 1.2: Survey result 1

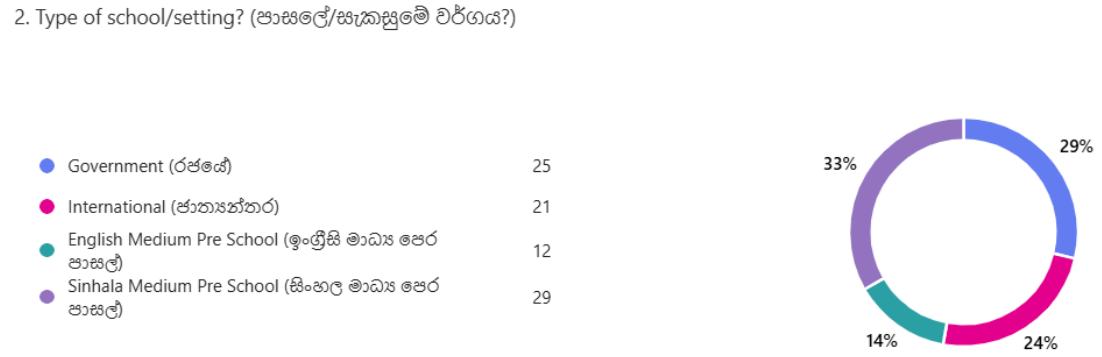

Figure 1.3:Survey result 2

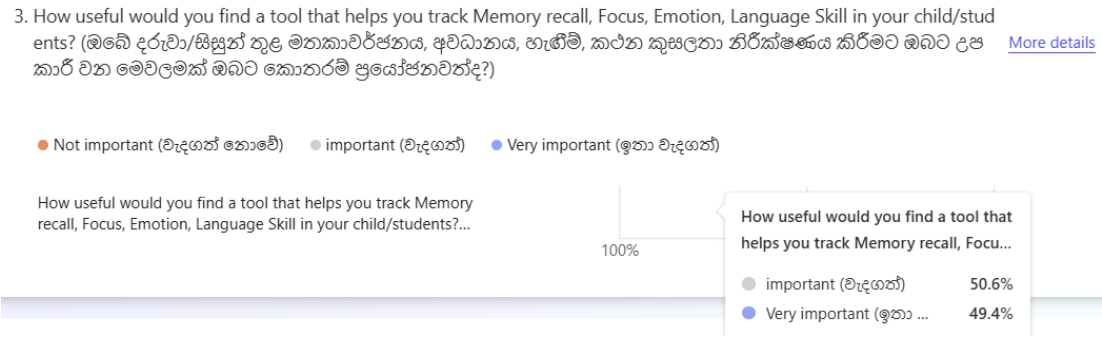

Figure 1.4: Survey result 3

4. How do you currently monitor Memory recall, Focus, Emotion, Language Skills? (ඔබ දැනට මතකාවර්ජනය, අවධාන ය, හැඟීම්, භාෂා කුසලතා කුසලතා නිරීක්ෂණය කරන්නේ කෙසේද?)

Figure 1.5: Survey result 4

5. Primary language used at home ? (නිවසේදී භාවිතා කරන ප්‍රාථමික භාෂාව ?)

Figure 1.6: Survey result 7

6. Do you use English as the secondary language? (ඔබ ද්විතියික භාෂාව ලෙස ඉංග්‍රීසි භාවිතා කරනවාද?)

Figure 1.7: Survey result 8

7. What activities does your child do most? (ඔබේ දරුවා වැඩිපුරම සිදුකරන ක්‍රියාකාරකම් මොනවාද?)

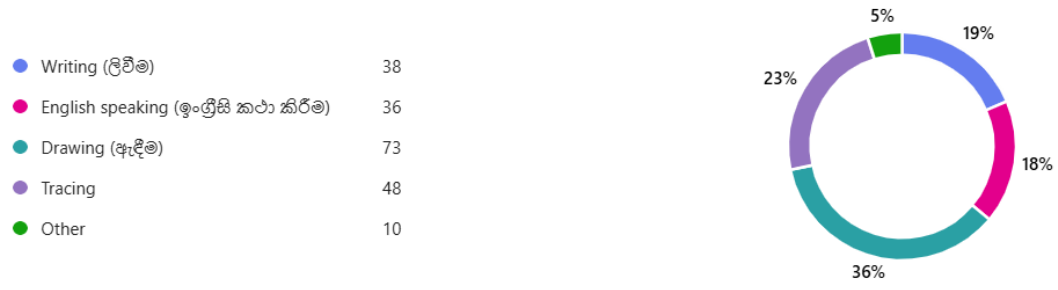

Figure 1.8: Survey result 9

8. How often does your child write letters at home ? (ඔබේ දරුවා නිවසේදී කොපමණ වාරයක් අකුරු ලියනවාද ?)

Figure 1.9: Survey result 10

9. Do you think an AI tool will be good for this? (මේ සඳහා AI මෙවලමක් හොඳ යැයි ඔබ සිතනවාද?)

Figure 1.10: Survey result 11

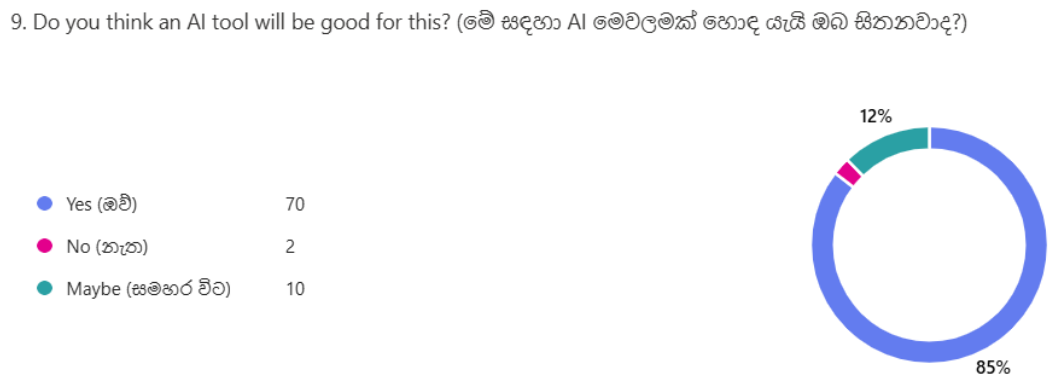

Figure 1.11: Survey result 12

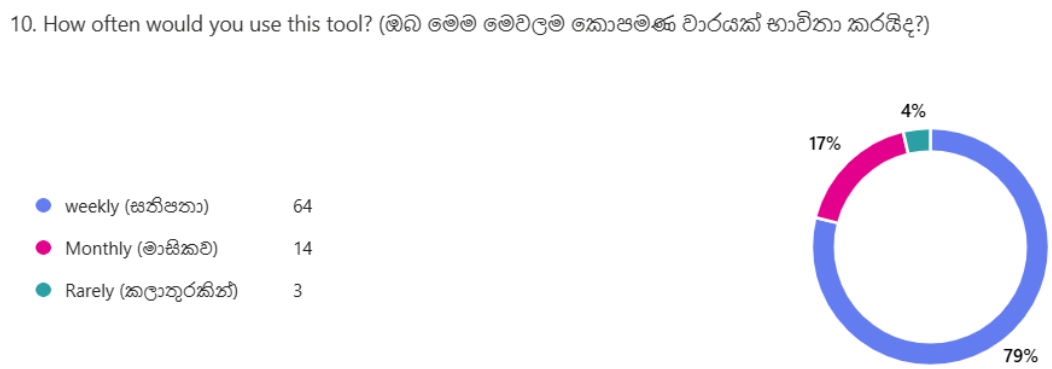

Figure 1.12: Survey result 13

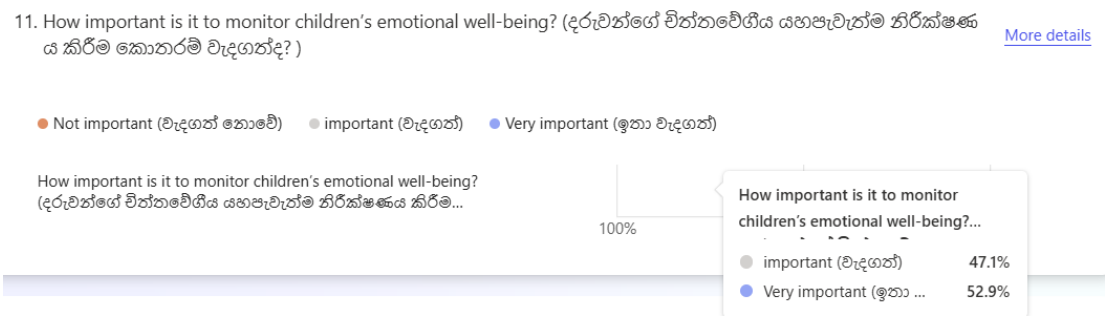

Figure 1.13:Survey result 14

This survey was filled mostly by parents whose children, go to Sinhala medium preschools, mostly speaking in Sinhala and used English as the secondary language. Most of the responders agreed that a tool that can track memory recall, focus, emotion and language skill will be useful. Currently no formal method is being used for these tasks. 85% of the people agreed that an AI tool would be helpful. Out of 87 people

who filled the form, 64 responded that this should be assessed in a weekly basis. Overall, every single respondent agreed that monitoring a child's emotional wellbeing is a crucial tack.

1.1.2. Literature Survey

Memory Recall in Early Childhood

Memory recall refers to accessing stored information for learning and problem solving, divided into short-term and long-term systems [14]. Cued recall is considered a more valid measure for preschoolers (aged 4-6) as it reduces retrieval burden [6]. Early childhood is marked by rapid hippocampal and prefrontal development, enabling children to remember rules, phonetics, and concepts. Strong recall skills support literacy, numeracy, and socio-emotional adaptation [10],[15].

Memory recall is also very crucial in the socio-emotional development. In addition to academics, children may have to retrieve coping strategies as taught by teachers. An example is a child who recalls a breathing exercise at the time s/he feels frustrated as such an act displays the functional unity of recall and EI in self-regulation [5]. In the same way, the positive rewards of sharing or turn-taking rhyme will aid in directing the behavior in a social situation. These examples demonstrate that recall cannot be discussed only as a separate cognitive ability but as a habit that is incorporated into the general system of emotional and behavioral adaptation.

Emotional regulation skills are very strong mediators of recall performance even in preschool settings. Emotional intelligence, especially the capability to control impulses and stimulate mood self-correction, directly depends on learning results. As children are better able to handle frustration, deal with delays, or stay calm as they become anxious, the more consistently they perform on a recall task in comparison with less easily dysregulated peers [5]. Emotional states strongly influence recall: positive affect improves memory, while stress and anxiety impair it by disrupting hippocampal processing [16].

Handwriting provides a sensitive measure of recall since it engages visual, phonological, and motor systems simultaneously. Unlike typing, handwriting activates

widespread brain networks and improves memory consolidation [8],[9],[16]. In bilingual settings, orthographic differences between Sinhala and English impose higher recall demands, strengthening both visual and phonological memory [10].

Categorization is another foundation of semantic memory. Cognitive psychology views categorization as playing key role in semantic memory since it shows how children categorize and store information into the long-term systems [12].

Review of Existing Approaches

Handwriting recognition has long been a central topic in pattern recognition and educational AI. While the field originally developed to support applications such as Optical Character Recognition (OCR), more recent research has emphasized its potential for educational use, including the assessment of memory recall in young learners. Early breakthroughs were made by Grave [17], who introduced a Multidimensional Recurrent Neural Network (MDRNN) coupled with Connectionist Temporal Classification (CTC), eliminating the need for manual segmentation and significantly improving recognition accuracy. Their model could handle unconstrained handwriting with irregular shapes and spacing, making it well-suited for analyzing the unpredictable and unrefined writing samples of preschoolers.

To address the challenges of noisy, irregular writing, Liu conducted a landmark benchmarking study of handwritten digit recognition techniques [18]. Their results showed that neural networks consistently outperformed classical statistical methods and template-based approaches, especially under noisy conditions. Such findings are directly relevant to preschool handwriting, where incomplete strokes and uneven spacing mirror noisy data. Neural networks introduced adaptability and robustness, laying the foundation for their application in assessing children's early writing.

The introduction of Convolutional Neural Networks (CNNs) by LeCun marked a major advancement [19]. CNNs are designed to extract hierarchical features starting with edges and curves in the early layers and progressing to complex structures in deeper layers. This multi-level feature learning enables the recognition of subtle variations in handwriting, such as distinguishing between correct and incorrect letter formations. For recall assessment, this fine-grained capability is essential because accurately reproducing letter forms reflects a child's memory quality. CNNs also

improved scalability, enabling large-scale handwriting assessment with minimal preprocessing.

Rosyda and Purboyo [20] further reviewed handwriting recognition methods, noting that CNNs surpassed classical machine learning models like k-nearest neighbors and multilayer perceptrons. Their review highlighted challenges in cross-linguistic recognition, especially for scripts with complex ligatures or diacritics, which require specialized preprocessing. This observation is particularly important in bilingual contexts such as Sinhala and English, where preschool recall assessment must address fundamentally different orthographies.

In Sinhala handwriting recognition, challenges are even greater due to the script's rounded alphasyllabary structure, compound ligatures, and numerous diacritics. Unlike English, Sinhala characters often share structural similarities, complicating segmentation and recognition. Silva emphasized two major barriers: the inherent complexity of the script and the scarcity of annotated handwriting datasets [21]. These problems are amplified in preschool contexts, where handwriting is especially inconsistent.

Despite these obstacles, CNN-based methods have achieved promising results. Wasalthilake and Kartheeswaran developed a CNN model for Sinhala handwritten character recognition that achieved 95% training accuracy and 85.7% testing accuracy across 55 characters [22]. Their dataset contained 1,815 images collected from 33 writers aged 13-75, which were scanned and normalized for training. Notably, their experiments showed that small design changes such as adding pooling layers at early stages improved performance. Such insights are particularly relevant when extending recognition to preschool handwriting, where irregular strokes and distortions are common.

Complementing this, Mariyathas conducted a broader evaluation of CNN-based Sinhala handwriting recognition and reviewed different preprocessing and architectural strategies [23]. Their study provided a structured comparison of recognition methods and highlighted both opportunities and challenges specific to Sinhala. They emphasized data augmentation and transfer learning as pathways to address dataset scarcity, making their work an important resource for preschool recall

assessment research. Together, these studies establish a foundation for adapting handwriting recognition to bilingual early education.

General machine learning evaluations have reinforced the importance of architectural and training choices in recognition performance. Peng [24] compared CNNs with traditional classifiers like random forests and SVMs, concluding that while CNNs excel in extracting spatial features, hybrid and ensemble approaches remain useful in low-resource contexts. For preschool handwriting, where datasets are small, combining CNNs with transfer learning or feature-based methods may provide resilience. Gupta [25] compared OCR with large language model (LLM) based approaches and demonstrated that LLMs offered superior adaptability, handling noisy and irregular handwriting more effectively than CNN-only pipelines. This is particularly relevant in early recall tasks where preschoolers' letter forms often deviate significantly from standard shapes.

Beyond recognition, handwriting has also been studied as a window into cognitive and motor performance. Vanjari and Shete [26] developed a CNN-based system to predict cognitive impairment using handwriting samples. Their model captured stroke-level motor irregularities beyond mere character identity, showing how handwriting analysis can assess broader developmental patterns. This reinforces the role of handwriting recognition in early childhood, where recall assessment requires not only transcription accuracy but also evaluation of motor planning and cognitive processes.

Puranik examined the dimensionality of letter-writing ability among preschoolers and found it to be a unidimensional but unevenly acquired skill [27]. Some letters were consistently written correctly earlier, while others took much longer to master. These results highlight the need for adaptive scoring systems that can provide nuanced feedback reflecting developmental progression rather than binary outcomes.

Parallel to handwriting recognition, categorization tasks play a central role in assessing semantic recall. Fu [28] studied categorization abilities in children with moderate intellectual disabilities in China, showing that their recall and learning were significantly weaker than those of typically developing peers. This highlights categorization as a sensitive measure of memory processes in early childhood.

Heit [29] modeled category learning and demonstrated that prior knowledge influences recall performance, with children better able to categorize and remember items consistent with their existing schemas. Such findings underscore the technical importance of building categorization systems that can measure how preschoolers group objects in recall tasks.

Datasets have been instrumental in supporting categorization research. Deng [30] introduced ImageNet, a large-scale dataset with millions of labeled images across thousands of categories. ImageNet transformed object detection algorithms but contains mostly adult-oriented categories, limiting its direct relevance to preschool recall assessment. To address this, Mureșan and Oltean [31] created the Fruits-360 dataset, designed specifically for fruit and vegetable recognition. Its clean backgrounds and familiar categories make it highly appropriate for preschool recall tasks, where children often engage in sorting exercises using foods or toys. Training models on such age-appropriate datasets allows recall assessment systems to automatically evaluate whether a child has categorized objects correctly, reducing the need for manual teacher scoring.

Object detection algorithms themselves have evolved significantly, providing technical infrastructure for categorization recall assessment. John and Meva [32] compared object detection algorithms, highlighting the limitations of sliding-window methods and the advantages of deep learning-based detectors such as YOLO and SSD, which achieved higher efficiency and accuracy. These real-time detectors align with classroom needs, where immediate feedback is essential during recall activities.

Amjoud and Amrouch [33] reviewed CNN-based and Vision Transformer (ViT)-based object detectors, emphasizing that transformers capture global relationships and semantic context across images. Such models mirror the conceptual reasoning in categorization recall tasks, where preschoolers must group items based on semantic categories like “animals” or “fruits,” not just visual similarity. Together, these comparative and review studies confirm that object detection algorithms can be adapted to preschool recall contexts by aligning technical categorization with cognitive recall frameworks.

Radford [34] introduced the Contrastive Language-Image Pre-training (CLIP) model, a vision language system trained on 400 million image-text pairs. CLIP supports natural-language-based categorization, allowing preschool memory tasks to be dynamically generated using teacher-defined prompts such as “things you eat” or “red objects.” This greatly increases the range of memory recall activities that can be scored automatically without retraining the model. CLIP can recognize visual-semantic relationships even when explicit category labels are missing, making it highly adaptive for early learning environments where recall exercises often vary week to week.

1.2. Research Gap

The memory recall in early childhood has been researched in diverse fields, yet a more detailed inspection of the currently available tools allows identifying prominent gaps. Most of the existing methods are technology-dependent, e.g., handwriting recognition using tablet devices and categorization applications on mobile devices, leading to an excess of screen time among young children. Developmental experts warn that spending too much time in front of screens at an early age limits attention span, poses a threat of being digitally addicted, and has a harmful effect on a child’s social-emotional progression [13]. Although these technologies are convenient, they do not match with the best practices of such preschool-aged children.

Secondly, most of the current instruments of recall assessment do not emphasize on recall accuracy but on attention span and processing speed or motor skills. As an example, handwriting instruments may measure accuracy of pen work or motor control instead of stating that there is a good recall of a letter [27]. This ignores the thinking element of recalling, which is the most important aspect in learning readiness of children and their emotional skill intelligence.

Thirdly, the use of a multimodal approach is very rare with recall assessments. Most systems distance one of the writing and categorization tasks instead, without combining them. However, research indicates that in order to ensure that the assessment of cognitive development is of reliable nature, the use of multimodal recall tasks would be imperative as they involve multiple modalities of memory: visual-

linguistic memory engaged in handwriting and semantic memory engaged in categorization [35]. Combined application of both modalities is yet to be widely used in conventional assessments.

Another restriction is that of lack of bilingual recall tasks which reflect to a reality of countries like Sri Lanka. In moved up locations, children grow up being exposed to both Sinhala and English at an early age of schooling. Some recall tools that are in existence currently are mainly developed to cater to the English speaking and the other alphabetic languages [10]. Sinhala is distinct in the sense that the complex visual forms of an alphasyllarabary must be reconciled with poor mnemonic characteristics of the language. Henceforth, a dual language examination that looks at recall in both Sinhala and English while recording would afford more information to the memory recall in dual language situations.

Table 1.1: Comparison of existing research with proposed system

Research Gap	**Research A [17]**	**Research B [29]**	**Research C [27]**	**Research D [28]**	**Proposed System**
Focus on Recall Accuracy	✗	✗	✓ (motor) focus)	✗	✓
Uses Handwriting Recall	✓ (digital input)	✗	✓	✗	✓ (paper → AI analysis)
Uses Categorization Recall	✗	✗	✗	✓	✓
Bilingual Recall (Sinhala + English)	✗	✗	✗	✗	✓
Screen-Free Approach	✗	✗	✗	✗	✓
EI Integration (self-regulation link)	✗	✗	✗	✗	✓
Suitable for Sri Lankan Urban Preschool Context	✗	✗	✗	✗	✓

Why These Gaps Matter

It is evident by overdependence on digital screens that there is a discrepancy between the technology tool and developmental demands. Whereas in higher-income settings mobile and tablet-based tests are fast becoming common, these are inappropriate in children aged between 4 and 6 years, who respond better to touch and the reality of the world [8]. This is very worrying especially in Sri Lanka where parents and teachers are becoming worried about the overexposure of children to mobile devices [9].

The motor performance requirement over the accuracy of recall does not reflect the essence of memory retrieval. Measures such as stroke smoothness or writing speed cannot reliably indicate memory ability, as they do not assess whether children can correctly recall and reproduce target information. This limitation may lead to an incomplete understanding of memory development and overlook children who are at risk of memorization-related learning difficulties.

In the same spirit, the lack of multimodal testing restricts the diagnostics. Deliberations on cognitive psychology research have supported the use of multiple testing modalities of the memory system (visual, linguistic, semantics recall) as a more valid indicant of cognitive resilience [35]. This proposal intends to fill this methodological gap, by combining both handwriting and categorizing.

Lastly, non-existence of bilingual assessment tools is another feature that lowers the situation in Sri Lanka. The majority of assessment available designs follow English speaking education system where the scripts are alphabetic. Sinhala does have complex curves and ligatures, as well as many strokes to a grapheme which adds an extra burden to recall. Such disregard of complexity leads to culturally and linguistically biased forms of assessment that do not replicate the reality of how children experience education in Sri Lanka [21].

Contribution of this study

This study fills the gaps that have been identified by making four important contributions to the current body of knowledge in the following ways:

1. Non-screen assessment of recall: There is an assessment of the recall, but it is done on paper and is only uploaded by teachers, thus children are not subjected to direct digital exposure as a result.
2. Emphasis of recall accuracy: AI analysis judges the accuracy of the reproduction of the letters and classification of the objects, providing the assessment can be based only on a cognitive level of recall and not on motoring ability.
3. Multimodal implementation: Handwriting requires visual-linguistic memory, but categorization tasks also rely on semantic memory, providing a broader picture of memory.

The four gaps that the proposed memory recall module bridges are an improvement in the field of early childhood monitoring of EI. It does not only add a new bilingual-multimodal framework, but it is also developmentally appropriate, as it avoids the screen-dependency.

1.3. Research Problem

The early childhood memory recall has long been explored, yet there are major weak points in its current research when it comes to preschool children, especially in bilingual environments where young learners are, like in Sri Lanka. Another crucial matter is the excessive use of digital platforms to estimate recall. Tablet-based handwriting and mobile classification software and categorization tools are gaining traction due to the convenience and auto grading features they allow. Nonetheless, developmental psychologists warn that over exposure to screens in early childhood can have negative effects on containers such as a shortened attention span, decreased emotional controls and a greater risk of becoming dependent on digitization [13]. Children aged 4 years to 6 years have developmental best practices that favor tactile based, play based, and real-life interactions over prolonged interactions on digital media. This forms a mismatch between the available assessment tool and developmental needs of the children.

The other limitation is the fact that the lies a lot of recall assessment tools typically assess a proxy to memory as opposed to memory recall itself. Delayed examples are the handwriting applications that usually estimate the efficiency of a penstroke or writing speed or motor functionality [27]. Although these are effective in fine-motor development, it does not necessarily verify that a child has effectively recalled and re-created the desired information. This has resulted in most of the current tools neglecting the cognitive aspect of recall which is of utmost importance in learning readiness and the emotional intelligence process. The children who are weak in recall but have a developed motor ability will not be recognized as being at risk due to memory-based complications.

Also, the majority of the tools implement a single-modality method, either applying to handwriting or using the categorization tasks. Indeed, studies in cognitive psychology indicate that multimodal (i.e. combining visual-linguistic recall with handwriting and semantic recall with categorization) persuasion helps form a more complete and reliable assessment of memory performance in children [34]. In its absence, the current systems cannot support the variabilities in storage and access of memories in the early childhood.

Lack of bilingual recall tests also represents a very big gap. In the Sri Lankan cities, the children are exposed to both Sinhala and English at the preschool opposed to one language at a time. The scripts have a fundamental difference in orthographic complexity: Sinhala characters can be complex where they have to be reproduced using curvy letters with multiple strokes, whereas English letters are linear and simple [10],[21]. Recent recall tools have been almost entirely designed to work in English or other alphabetic languages and take no consideration of the constraints that alphasyllabaries such as Sinhala present. Such lack of bilingual reinforcement not only compromises the effectiveness of recall measurement in Sri Lanka but also the danger of cultural bias on the developmental assessment.

The limitations therefore require a recall assessment system that:

- Naturally and developmentally appropriate activities including handwriting and object categorization recall relative accuracy directly.

- Reduces the direct exposure screen time on children with digital technology limited to uploads by educators and Artificial Intelligence analysis.
- Includes multimodal recall tasks that measure both visual-linguistic and semantic memories.
- Takes the form of a bilingual situation as it measures recall in both Sinhala and English to represent genuine learning in preschools in Sri Lanka.

In short, the key research question is that the existing tools to assess recall cannot meet developmental, methodological, contextual needs of early childhood in the bilingual environment. They are too reliant on screens, have an emphasis on motor rather than memory retrieval, do not take into consideration the idea of multimodal integration, and cannot be utilized bilinguistically. This research aims to catalogue such a gap through the design and testing of a non-screen, multimodal, bilingual recall assess module which can be incorporated into an AI-driven emotional intelligence monitoring system, with a specific focus on the emotional intelligence construct of self-regulation.

2. RESEARCH OBJECTIVES

2.1.Main Objective

The main objective of this research is to help parents and teachers to monitor child's emotional intelligence development and find out whether any critical situations have occurred.

2.2.Sub-Objectives

1. Develop handwriting recall tasks in Sinhala and English.
   Handwriting recall tasks will involve asking children to reproduce Sinhala and English letters that are presented to them using related images. This goal talks about the unavailability of bilingual recall testing instruments in the existing systems. Arguably, Sinhala presents a visualized complex alphasyllabary and English is an alphabetic script, thereby, a more stringent test of memory retrieval of the children would involve testing both languages. The bilingual format resembles the true language variety in the urban areas of Sri Lanka and guarantees the measures used in assessing recall is in a culturally and educationally consistent way [10].

2. Design categorization-based recall tasks for preschoolers.
   Inductive tasks will entail sorting (e.g., sorting fruits and vegetables, land and flying animals). This aim is needed because of the necessity of multimodal recall evaluation because of the consideration of semantic memory recall. Categorization is done as an addition to handwriting to assess how children categorize and recall conceptual knowledge, which is strongly associated with language acquisition and subsequently academic achievement [11].

3. Measure recall accuracy through AI-based analysis of uploaded task images.
   When completed tasks will be uploaded by teachers, the analysis of their accuracy will be conducted by AI models. In the case of handwriting, it will compare it in terms of accuracy of letters written, on the categorization, it will compare how accurate the grouping is. This goal makes the assessment objective and scalable, which is not the case of traditional teacher-led test, as it is both subjective and

inconsistent. Crucially the AI is used behind the scenes so that the child will not see the screen.

4. Incorporate background survey data to refine scoring weights.
   Performance in recall on children is a factor of both individual cognitive abilities and environmental and socio-cultural surrounds. This goal will be combined with parental background surveys that will show the environment at home including activities on learning, what languages were being heard and the socio-economic background. Integration of this data will make the scoring system environmentally sensitive and the risk of misunderstanding the performance that may be environmentally shaped rather than cognitively based decreases.

5. Provide teachers with a monitoring dashboard and parents with progress visibility.
   The ultimate goal is to create a dashboard that summarizes information into actionable knowledge. This will provide teachers with in-depth analytics to guide classroom planning and bring to parents simplified progress reports. This capability directly contributes to EI development due to early intervention (and possibility of personalized support), as well as enhanced cooperation between the schools and families [7]. Moreover, by rendering the results visible, it increases responsibility and promotes regular tracking of self-regulation skills of children.

3. METHODOLOGY

3.1.Overall System Methodology

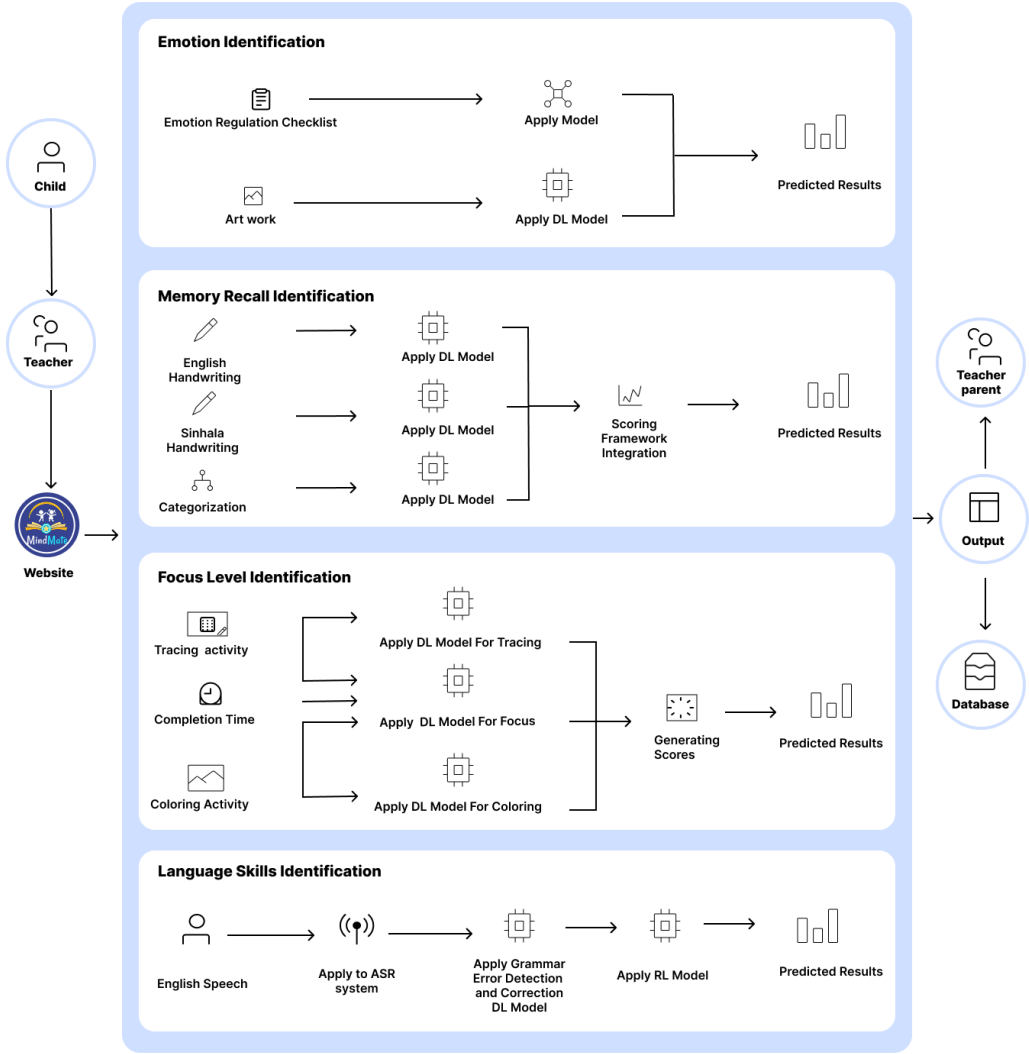

Figure 3.1: Overall System High Level Diagram

As illustrated in Figure 3.1, the system overview provides a comprehensive representation of a complete overview of the structure of the entire system. The system leverages advanced technologies, including image processing, audio processing, text processing, reinforcement learning, machine learning, deep learning and multi-model architectures, to analyze students' performance through cognitive skills, motor skills, emotions and language development. The system operates by collecting data through user interactions, where students engage with conversational voice assistants and image inputs of children's handwriting, categorization activities, drawings, behavioral

activities and line activities captured by teachers for analysis. The system then processes this data, extracting relevant features using sophisticated algorithms. A multimodal architecture is employed to integrate these features and make predictions regarding the student's performance across various areas such as emotion, focus level, memory recall and language skill which can assess Emotional Intelligence Pillars such as self-awareness, self-regulation, social skills, motivation and empathy. Feedback is subsequently provided based on the analysis in a teacher parent dashboard, which keeps the parent and teacher connected. The system leverages REST API Application Program Interfaces to facilitate seamless communication between different components, ensuring efficient data flow and real-time feedback. This holistic approach enables to get personalized feedback of children for teachers and parents.

3.2.Component Methodology

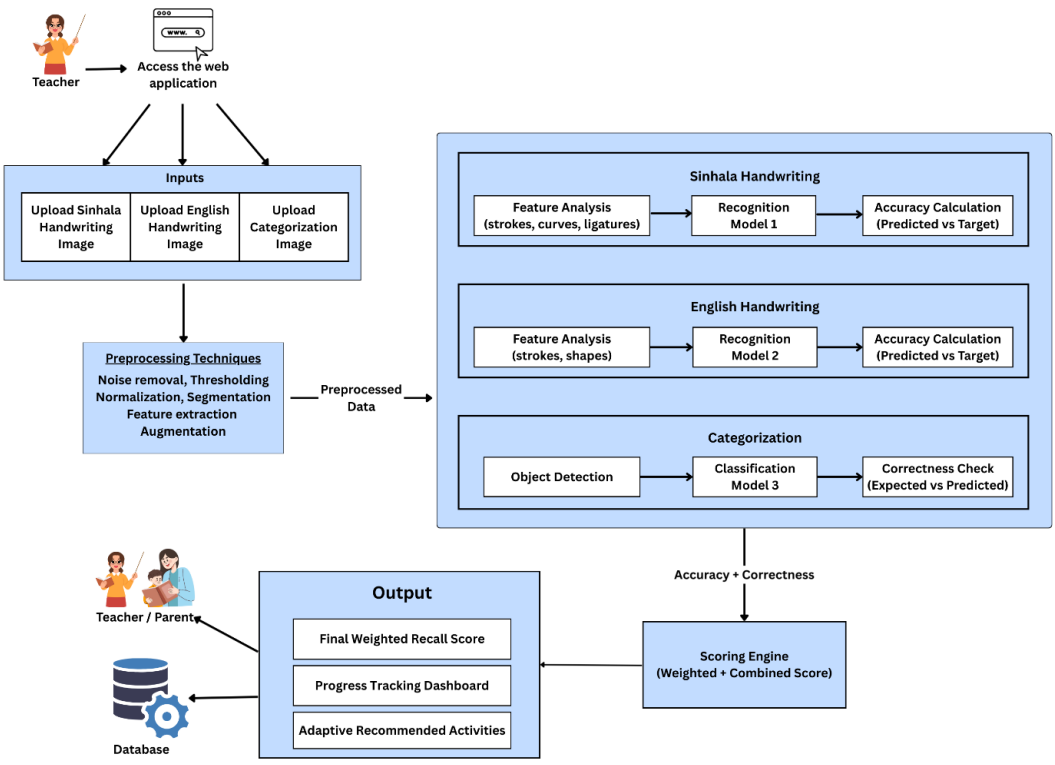

Figure 3.2: Component Diagram

3.2.1. Handwriting Task

Teachers will present an image of an object (e.g., an apple or a ball) to the child, who is then asked to write its first letter in Sinhala or English on the given sheet. After the child completes the handwriting exercise, the teacher saves a copy of the written sheet in the system by using either a mobile device or scanning the item.

The image uploaded will be processed using the AI algorithm, which matches the handwritten letters with the template forms of Sinhala and English scripts. The scoring framework comprises:

- Shape similarity - evaluates how closely the written strokes and curves align with the structural features of the intended letter
- Proportional accuracy - assesses the size, direction, and placement of the strokes relative to the expected form.

This activity directly assesses memory remembrance, as the success of the child is actually related to which of the two was retrieved, not how well they could write or copy the work.

3.2.2. Categorization Task

In this activity, the teacher presents some classifications: land animals and flying animals, fruits and vegetables, or shapes and colors. The child is then given a short pause after which he/she is told to sort them into similar categories using flashcards, blocks, or printed photographs.

After completion, the teacher captures the final arrangement as an image and uploads it to the system. The AI model then evaluates the submission by:

- Object Detection - identifying the items in the image (e.g., apple, carrot, cow, bird).
- Category Classification - assigning each item to the appropriate class using a trained model.
- Correctness Check - comparing the child's arrangement with the predefined answer sets to calculate accuracy.

3.2.3. Statistical Weighting of Tasks

Since handwriting and categorization are measures of different aspects of memory recall, statistical weighting will be used to overcome imbalances where weights will be used to generate comparable scores.

1. Pilot Study - Empirical testing with teachers and child psychologists will determine the relative importance of handwriting and categorization activities in preschool learning contexts.
2. Family Background Survey - Parental involvement, language exposure at home, and socio-economic factors will be gathered. These parameters will be employed to adjust the scores relating to handwriting in order to bring fairness and context responsiveness.

This combination of weighting systems makes it possible to guarantee that the assessment involves both the task fitness and environmental response.

3.2.4. Data Gathering

To support handwriting recognition and categorization activities, separate custom datasets will be developed.

1. Handwriting Data

Two handwritten datasets will be created:

- English handwriting dataset: This will include the letters J, F, M (uppercase) and m, f, j (lowercase).
- Sinhala handwriting dataset: This will include the characters ශ්‍ (ra), ද (u), ධ (ta), ල (la), ඩ (da).

The selected letters represent a mix of straight-line, curved, and compound strokes, making them suitable for evaluating recognition models while remaining manageable for preschool children. Each dataset will be built using samples from children aged 4-6 years, with multiple reproductions per letter. Data will be gathered through scanned sheets or high-resolution photographs, and contributions from multiple children will ensure the datasets capture natural variation in handwriting style.

2. Categorization Data

For the categorization activity, two complementary datasets will be developed:

- Custom object dataset: A curated collection of images such as fruits, vegetables, and animals will be sourced from open repositories and publicly available resources. Images will be preprocessed and augmented (e.g., rotation, scaling, and background variation) to enhance diversity. This dataset will be primarily used for training and validation.

- Children's completed categorization tasks: Captured images of children's sorted activities (e.g., grouping fruits vs. vegetables with picture cards) will be collected. These samples reflect authentic classroom conditions, including irregular placements and environmental noise, and will be mainly reserved for the testing phase, ensuring that model evaluation aligns with real-world preschool use cases.

3. Usage for Evaluation

The handwriting and categorization datasets will be divided into training, validation, and testing subsets. Controlled datasets (handwritten sheets and curated object images) will serve for training and validation, while children's completed categorization tasks will be reserved for the testing phase to evaluate practical applicability in preschool environments.

3.3. Research Constraints

3.3.1. Limited Availability of High-Quality Datasets for Sinhala Handwriting

Lack of digitized basics of Sinhala handwriting dataset is also one of the biggest challenges in this study. The majority of the work on handwriting recognition has been on English or other widely known scripts. Sinhala is an alphasyllabary with complicated shapes and ligatures that are especially difficult to analyze with regard to recall. As training AI models requires high-quality training data, the necessity to create a custom dataset is also likely in the course of this research. It will be labor-intensive

because it will involve acquisition and marking of handwriting samples of involved children.

3.3.2. Accuracy and Generalization of Models

The performance of handwriting recognition and categorization models depends heavily on the dataset's diversity. If the dataset lacks representation across different children, writing styles, or object categories, models may struggle to generalize. Moreover, achieving high accuracy with preschool data is inherently difficult due to the natural variability in children's outputs.

3.3.3. Data Privacy and Ethical Concerns

Ensuring proper consent from parents or guardians when collecting handwriting samples from children and protecting the privacy and confidentiality of children's data, especially when handling sensitive information.

3.3.4. Resource and Time Constraints

Training handwriting and image recognition models requires Graphics Processing Unit (GPU) resources, which may be limited. In addition, data collection, annotation, model training, and evaluation must all be completed within a short research timeline, where limited time may restrict iterative fine-tuning and reduce the opportunity to explore multiple model versions.

4. PROJECT REQUIREMENTS

4.1. Functional Requirements

1. Recall analysis of Handwriting
    - The system supports uploading children's handwriting images.
    - The system will grade handwritten Letters of children in Sinhala and English with a high level of accuracy.
    - The system shall output both numeric results (e.g., percentage) and descriptive labels (Correct / Partially Correct / Incorrect).

2. Recall analysis of Categorization
    - The system will allow the teacher to select a predefined category set and assign the session to a child.
    - The system will accept a single photo (or short clip) capturing the final arrangement of items.
    - The system will detect the items present and determine their placed category area.
    - The system will compare each item's placement to the reference category mapping and compute correctness per item and overall.
    - The system will present results as numeric scores and descriptive labels (Correct / Partially Correct / Incorrect).

3. Teacher/Parent Dashboard
    - The system will provide dashboards for teachers and parents to view child's recall performance, progress trends, and insights.
    - The system will generate personalized recommendations based on each child's performance.

4. Stored Data Security
    - Store all activity data securely in a database.
    - Support progress tracking longitudinally.

5. Authentication & Access Control
    - Teachers and parents must be able to log in with child-linked credentials.
    - Role-based access: Teachers can create and conduct recall sessions; parents can only view their child's progress and feedback.

4.2. User Requirements

1. The teachers are supposed to be able to:
    - Log in with secure offering.
    - Upload children's task sheets with minimal effort.
    - Receive highly detailed reports and analytics to inform interventions.

2. Parents should be able to:
    - View progress reports via portal.
    - Track their child's improvement over time.

4.3. Non-Functional Requirements

- Usability: The portal and dashboards used to upload should be easy to use and only minimal training is required.
- Reliability: The AI engine will provide similar scores in repeated entry as and when similar handwriting and paper quality are used.
- Performance: The task analysis should be done within a second of upload.
- Scalability: The existing system should be able to perform horizontal scaling to handle a greater number of children.
- Security: The user data must be held in a secure and only those with the rights should be able to access such data.
- Maintainability: The system will permit upgrade without difficulty.

4.4. Technical Requirements

1. Handwriting Recall Analysis

Available Technologies

- Traditional Approaches (OCR) - Open-source, lightweight, supports basic text extraction. But poor accuracy for preschool handwriting and non-Latin scripts.
- API-driven Services (e.g. Google Vision API) - High accuracy, multilingual support. But commercial cost per API call is high, not tailored for preschool handwriting.
- Advanced Deep Learning Techniques (e.g. CNN-based Models) - Can be trained on custom Sinhala + English preschool datasets, robust to noisy inputs. But it requires dataset preparation and GPU training.
- Transformer-based Models (e.g. Hugging Face CRNN, ViT) - Pretrained models for handwriting recognition, adaptable via fine-tuning. Fine-tuning requires annotated preschool data.

2. Categorization Recall Analysis

Available Technologies

- Classical Image Processing (e.g. OpenCV) - Lightweight, useful for detecting object positions. But limited for complex categorization tasks.
- Anchor-based Detectors (e.g. YOLO) - Single-stage detectors that use predefined anchors, efficient and robust for identifying multiple objects in real time. Open-source object detection framework. Require labeled training data.
- Two-stage Models (e.g. Faster R-CNN) - Region-proposal based detectors with high accuracy for complex categories, but slower and computationally heavier.
- Transformer-based Detectors (e.g. DETR, Vision Transformers) - End-to-end detection using attention mechanisms, pretrained on large datasets and highly adaptable. Require fine-tuning.

- Google AutoML Vision - Easy training, high accuracy. But commercial cost and vendor lock-in.

3. Dashboard & Progress Tracking

Available Technologies

- Frontend: React, Angular, Vue.js

  React chosen for reusability, ecosystem support, and smooth integration with visualization libraries.
- Backend: Flask (Python) vs Django (Python) vs Spring Boot (Java)

  Flask chosen for lightweight integration with AI models.
- Database: PostgreSQL vs MySQL vs MongoDB

  PostgreSQL chosen for structured relational data (users, recall scores, recommendations).

Selected Technology:

Stack: Frontend: React.js, Backend: Flask (Python), Database: PostgreSQL. Python backend aligns with AI/ML workflows; React provides smooth dashboards; PostgreSQL ensures reliable structured storage.

4.5. Expected Test Cases

Table 4.2: Expected test cases

Test Case ID	Test Case Description	Precondition	Expected Outcome
TC001	User registration	User opens the application	User account is successfully created and stored in the DB
TC002	Ensure user login functionality and secure authentication	User has a registered account in the database	User can log in securely with correct credentials, session created
TC003	Upload images of Sinhala handwriting.	Teacher logs from the registered account	Valid image uploaded and stored successfully
TC004	Upload images of English handwriting.	Teacher logs from the registered account	Valid image uploaded and stored successfully
TC005	Upload categorization image/clip	Teacher logs from the registered account	Media validated and uploaded successfully
TC006	Analyze all tasks	Teacher logs from the registered account	Shows the correct performance summary.
TC007	Validate secure logout and session management	Teacher logs from the registered account	User is securely logged out and session data is cleared

5. COMMERCIALIZATION

Customer value

Through the survey results we found out, 85% of the people would like to see an AI tool to monitor emotional intelligence. They also highlighted how important emotional intelligence monitoring is for early childhood development. The target audience for this application will be parents and teachers. Considering these results below are some of the steps that will be considered to commercialize the application.

Competitive analysis

Similar applications on emotional intelligence without AI models have an average price range of $9.99 monthly. Applications that use AI models have an average price of $20 monthly subscription.

How does our application differ from the competitors?

Exiting applications	Our application
Gamified applications that increase digital addictions of the children	Uploading the work the child has done to reduce digital addiction
Facial expression capturing to identify emotions	Does not capture or store children's facial expressions or biometric identifiers
Use activities to improve child specific skill area.	Use activities not only to improve their skills but also to get insights on emotional intelligence based on those activities.
Limited parental involvement or communication	Facilitates and enhances parent-teacher interaction and collaboration
Focus more on cognitive, thinking, and emotions with reward-based motivation	Provides meaningful insights and personalized recommendations to parents and teachers based on the child's activity outputs
Primarily rely on digital input (typing, drag-and-drop, touch-based tasks)	Ensures that no direct digital input is involved in children's activities.
Generic design without consideration for cultural or linguistic diversity	Incorporates both Sinhala and English, ensuring cultural and linguistic appropriateness.

User benefits of our application:

- Monitor child's emotional intelligence using multiple methods
- Builds emotional intelligence at an early age, which supports learning, social skills, and resilience.
- Helps teachers identify at-risk students early (e.g., children showing signs of sadness, anger, withdrawal)
- Teachers and parents both can help children to improve suggested activity skills based on dashboard feedback.

Cost analysis

On average a web hosting platform will cost $7 with Postgresql database support (eg: Hostinger). This will be the rough hosting cost of our application.

Pricing plan

Considering the above factors this application will have a one-month free trial period, which will be used to gain customers into the application. After the one-month trial, the user will have to pay $15 monthly subscription.

- Standard plan ($15) – One subscription per student.
- Family plan ($30) – One subscription per three students.
- School plan ($40) – Class registration ability, one subscription per 20 students.

6. GANTT CHART AND WORK BREAKDOWN STRUCTURE

6.1. Gantt chart

Task Name	June	July	Aug	Sep	Oct	Nov	Dec	Jan	Feb	Mar	Apr	May
Research topic selection												
Topic assesement												
Project charter												
Study on research area												
Project proposal report												
System design and planning												
Implementation of functions												
Integration level 1												
Testing level 1												
Progress presentation 1												
Check List 1												
Prepare research paper												
Research paper												
Implementation of functions												
Testing level 2												
Progress presentation 2												
Check List 2												
Final report												
Final presentation												

Figure 6.1: Gantt Chart

6.2. Work Breakdown Structure

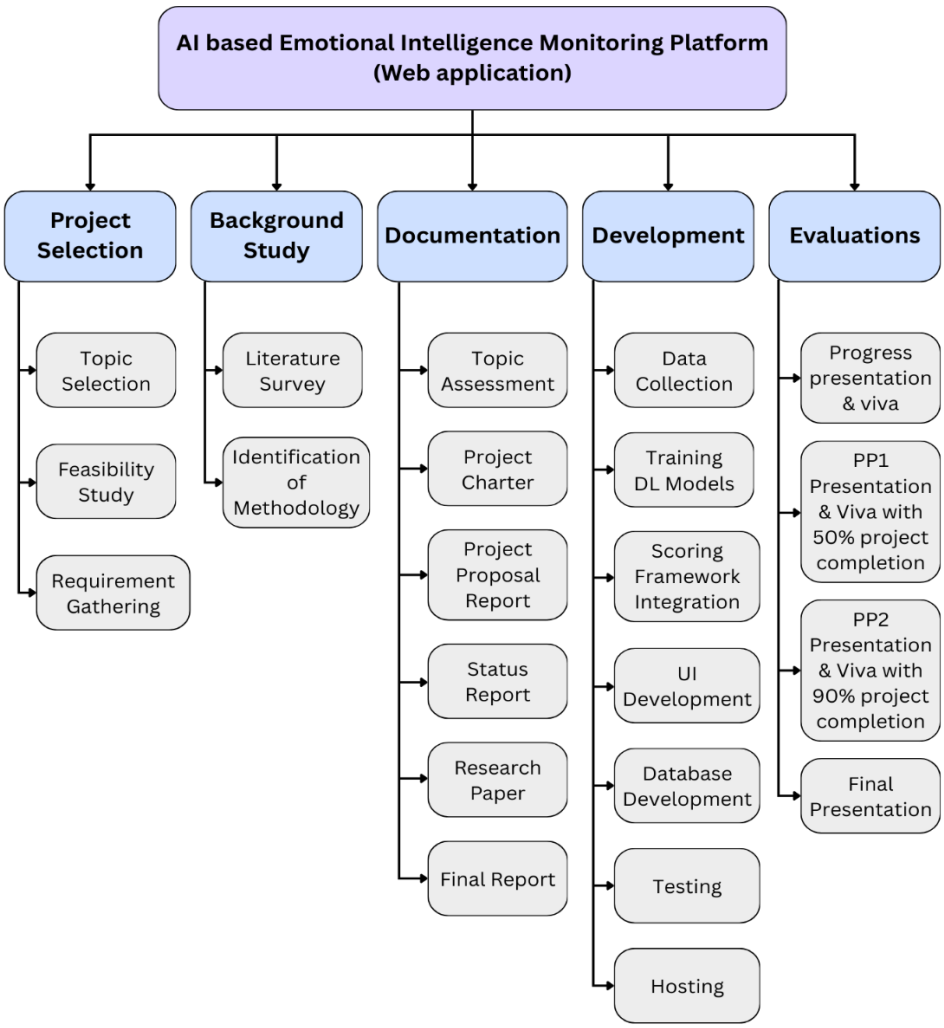

Figure 6.2: Work Breakdown Structure

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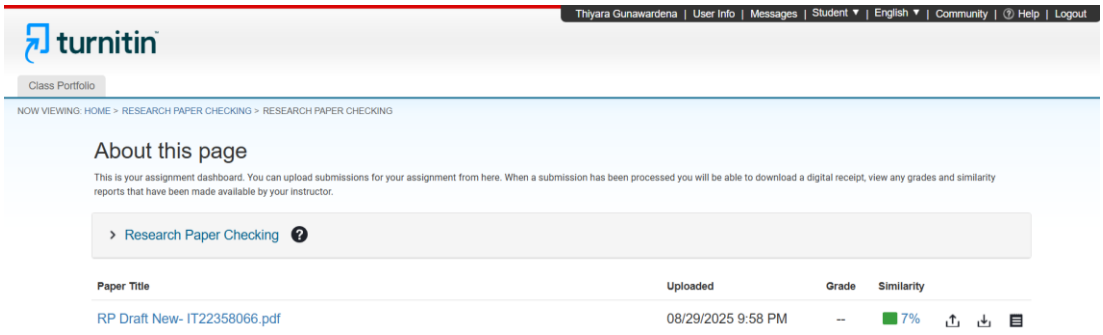
APPENDICES

Appendix - A: MindMate Logo

Appendix - B: Online Survey

https://forms.office.com/r/BVupx17rcQ

Appendix - C: Plagiarism Score using Turnitin